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Fluid Mechanics.

⇒ Important Terms.

1. Fluid : Anything that has the ability to flow.

Shaft → Mechanical Component to transmit torque
Pipe → " " used to " fluid

The study of fluid at rest is called Fluid Statics
Fluid Kinematics : Study of fluids in motion where Pressure forces are not considered
Fluid Dynamics : " " " " " " " " are considered.

Specific Weight / weight Density.
It is Wt. of the Fluid divided by volume of the fluid.

$$\text{Specific wt. or wt. Density} = \frac{\text{Wt. of Fluid}}{\text{Volume of "}}$$

⇒ Specific volume of a fluid : is defined as the volume of fluid occupied by a unit mass

$$\text{Specific Volume} = \frac{\text{Wt. of Fluid}}{\text{Mass " "}} = \frac{1}{\rho}$$

(4) Specific Gravity.
is defined as the wt density of fluid to a wt density of a standard fluid

$$\boxed{S(\text{for liquid}) = \frac{\text{Wt density of a fluid}}{\text{Wt " " water}}} = \frac{\rho_1 g}{\rho_2 g} = \frac{\rho_1}{\rho_2}$$

$$\boxed{S(\text{for gases}) = \frac{\text{wt. density of gas}}{\text{wt. density of air}}}$$

Q Calculate specific wt. density & specific gravity and 1L of fluid which weighs 7 kg./N.

Ans.

Volume = 1000 cm³, Wt. = 7 N

$$\text{Specific Weight} = \frac{\text{Wt.}}{\text{Volume}} = \frac{7\,N}{\left(\frac{1}{1000}\right)m^3} = 7000\ N/m^3 \rightarrow ①$$

$$\boxed{\text{Density} = \frac{\text{Specific wt.}}{g\,(9.81)}}$$ Use ① here

Density = ~~7~~ 714.28 kg/m³

⇒ $$\text{Specific Gravity} = \frac{\rho_{liq}}{\rho_{water}} = \frac{714.28}{1000}\,\frac{kg/m^3}{kg/m^3} = \_$$

④ Viscocity.

When the layers of a fluid at a distance dy apart move over one another at different velocities lets say u and u+du, the viscocity along with the relative velocity causes a Shear stress acting b/w the fluid layers.

Shear Stress ∝ Rate of change of Velocity

$$\tau \propto \frac{du}{dy}$$

$$\boxed{\tau = \mu \cdot \frac{du}{dy}}$$

Newton's Law of Viscocity

μ: Constant of Proportionality as dynamic viscocity

$\frac{du}{dy}$ = Rate of Shear Strain

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All liquids that obey the above law are called newtonian fluids

Viscosity: is defined as shear stress required to produce unit rate of shear strain.

$$\mu = \frac{\tau}{(du/dy)} = \frac{\text{Shear } \cancel{\text{Strain}}\ \text{Stress}}{\left(\frac{\text{Change in Velocity}}{\text{'' '' Distance}}\right)}$$

$$= \frac{\text{Force / Area}}{\frac{\text{Length}}{\text{Time}} \times \frac{1}{\text{length}}}$$

$$= \frac{\text{Force} \times \text{Time}}{(\text{length})^2}$$

In MKS System.

$$= \frac{kgf \times sec}{m^2}$$

CGS System $= \frac{dyne \times sec}{m^2}$ = Poise

S.I. units $= \frac{Ns}{m^2}$

$$1 \text{ dyne} = 1\,gm \times \frac{1\,cm}{s^2}$$

$$= 10^{-5}\ N$$

1 Centipoise $= \frac{1}{100}$ poise

$$1\ C.P. = \frac{1}{100}\ P.$$

Kinematic Viscosity $= \nu$ (nu) $= \frac{\mu}{\rho} \left(\frac{\text{Dynamic Viscosity}}{\text{Density}}\right) \Rightarrow m^2/s$ (MKS)

CGS (cm^2/s) = 1 Stoke

1 Centistoke $= \frac{1}{100}$ stoke

① For Liquids

$$\mu = \mu_0 \left(\frac{1}{1+\alpha t + \beta t^2} \right)$$

μ_0 = Viscocity of liquid at 0°C

μ = " " " " t°C

② For gas

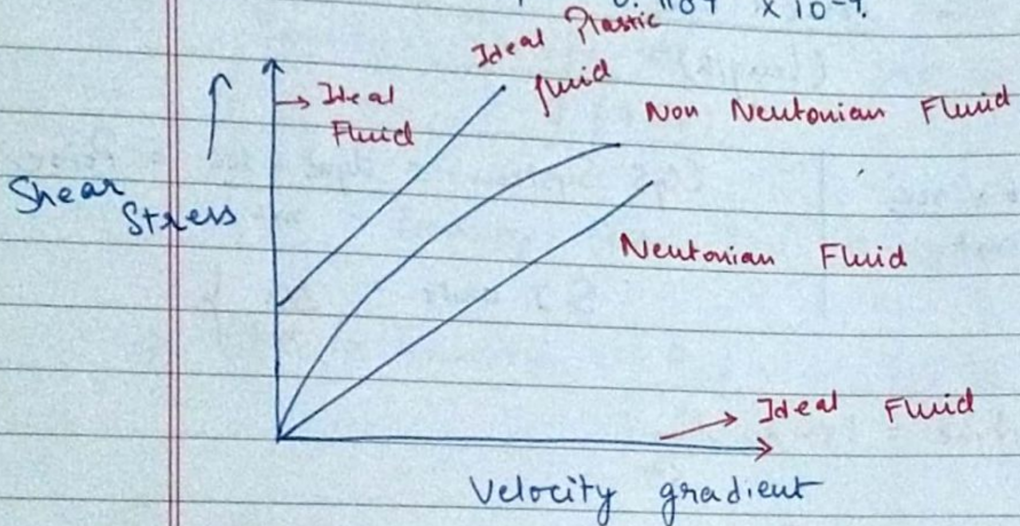
$$\mu = \mu_0 + \alpha t - \beta t^2$$

For air,

$\mu_0 = 0.000017$

$\alpha = 0.000000056$

$\beta = 0.1189 \times 10^{-9}$



Definition of Ideal Fluid

A fluid which is incompressible & is having no viscocity is called ideal fluid. Ideal fluid as all the fluids have some viscocity

" " Real Fluid

A fluid which possesses viscocity are called real fluids. All the fluids in actual practice are real fluids.

Ideal Plastic Fluid

A fluid in which shear stress is more than yield value

Shear Stress & Rate of Shear Strain

Q: If the velocity distribution over a plate is given by $u = \frac{2}{3}y - y^2$ in which u = Velocity in m/s at a distance y above the plate. Determine the Shear Stress at $y = 0$ & $y = 0.15$ m. Given μ = Dynamic Viscosity = 8.63 P.

Ans:

$$u = \frac{2}{3}y - y^2 \Rightarrow \frac{du}{dy} = \frac{2}{3} - 2y.$$

When: $\left(\frac{du}{dy}\right)_{y=0} = \frac{2}{3} \rightarrow ①$, $\left(\frac{du}{dy}\right)_{y=0.5} = \frac{2}{3} - \frac{2 \cdot 100^{-1}}{15^{-1}}$

$$= \frac{2}{3} - 0.3$$

$$= 0.66 - 0.3$$

$$= 0.33 \longrightarrow ②$$

⇒ Use $\tau = \mu \cdot \frac{du}{dy}$

(A) Isothermal Process

$$\frac{P}{\rho} = \text{constant}$$

Adiabatic Process

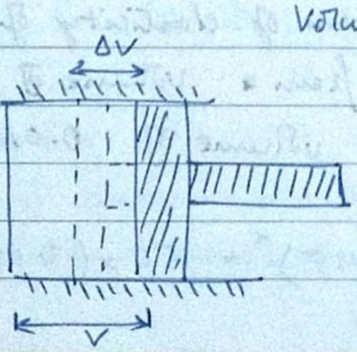
$$\frac{P}{\rho^K} = \text{Constant}, \quad K = C_P/C_V$$

K = Ratio of Specific Heats

$k = 1.4$ for air

⇒ Compressibility (Reciprocal of Bulk Modulus)

$$(\text{Bulk Modulus})_K = \frac{\text{Compressive Stress}}{\text{Volumetric Strain}}$$



$$\text{Volumetric Strain} = -\frac{dV}{V}$$

$K = \dfrac{\text{Increase in Pressure}}{\text{Volumetric Strain}}$

$$= \frac{dP}{-dv/v} = -\frac{dP \cdot v}{dv}$$

⇒ Isothermal Process:

$\dfrac{P}{\rho} =$ constant

$PV =$ Constant

Differentiating ⇒ $P\,dv + v \cdot dP = 0$

$$\Rightarrow P = -\frac{dP \cdot v}{dv} = K.$$

Adiabatic Process

$\dfrac{P}{\rho^k} =$ constant , $PV^k =$ Constant

$$\Rightarrow P \cdot \frac{d(v^k)}{dv} + v^k \, dP = 0$$

$$\Rightarrow P \cdot k \cdot v^{k-1} + v^k \, dP = 0$$

Cancelling v^{k-1}

$$\Rightarrow P \cdot k \, dv + v \cdot dP = 0$$

$$\Rightarrow P \cdot k = -\frac{V \cdot dP}{dv} = K \Rightarrow \boxed{P \cdot k = K} = \text{Ans.}$$

$K =$ Bulk Modulus, $k = \gamma$.

Q: Determine the bulk modulus of elasticity of liquid, which is compressed in a cylinder from a volume of $0.0125\,m^3$ at $80\,N/cm^2$ Pressure to a volume of $0.0124\,m^3$ at $150\,N/cm^2$

Ans: $P_1 V_1^k = P_2 V_2^k$

$$\Rightarrow 80 \times (0.0125)^k = 150 \times (0.0124)^k$$

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$$\Rightarrow \frac{8}{15} = \left(\frac{0.0124}{0.0125}\right)^k \Rightarrow \frac{8}{15} = \left(\frac{124}{125}\right)^k$$

$$\Rightarrow \log(8/15) = k \log\left(\frac{124}{125}\right) \Rightarrow -0.273 = k \cdot (-0.003)$$

$$k = \frac{273}{3} = 91.$$

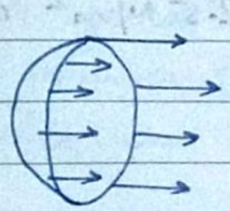
$$\Rightarrow \frac{8}{15} = \left(\frac{124}{125}\right)^k$$

$$\hookrightarrow \log\left(\frac{8}{15}\right) = k \cdot \log\left(\frac{124}{125}\right)$$

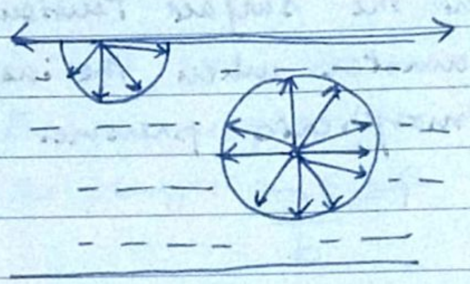
$$-0.273 = k \cdot (-0.0034)$$

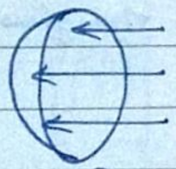
$$k = \frac{0.273}{0.0034} = 80.29.$$

→ Liquid Droplet



σ





P.

Via Force Eqⁿ

Tensile force = $\sigma \times$ Circumference

$\Rightarrow$ Pressure Force = $P \times \pi d^2/4$

$$\Rightarrow P \cdot \times \frac{\pi d^2}{4} = \sigma \times \pi d$$

$$\Rightarrow \boxed{P = \frac{4\sigma}{d}}$$

hollow Bubble

$$\frac{\pi d^2 \times}{4} P = 2(\sigma \times \pi d)$$

$$= \frac{8\sigma}{d}.$$

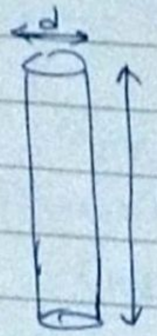
(σ) Surface Tension is defined as the tensile force acting on the surface of liquid in contact with the gas or on the surface b/w two immiscible liquids such that the contact surface behaves like a membrane under tension.

Immiscible liquids which cannot be mixed together

Units: σ in MKS =

in S.I = N/m

⇒ **Surface Tension in Liquid Jet**



Force due to pressure = Pressure × Area of Sur...

$= P \times L.d.$

Force due to Surface Tension $= \sigma \times 2L$

$\Rightarrow \quad P.L.d = \sigma . 2L$

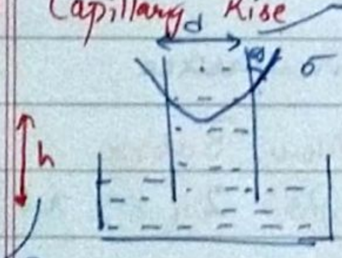
$\Rightarrow \boxed{P = \frac{2\sigma}{d}}$

Q: Find the surface tension in a soap bubble of 40mm diameter when the inside pressure is $2.5\ N/m^2$ above atmospheric pressure.

Ans. $P = \frac{8\sigma}{d}$.

θ b/w glass & Hg
128°

Capillary Rise → diameter of tube



d = diameter of tube

Weight of liquid of height h in tube

$= \left[(\text{Area of Tube}) \times h \times \rho \right] \times g$ ← Mass of Fluid

$= \frac{\pi d^2}{4} \times h \times \rho \times g$

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Intensity of Pressure $\simeq$ Pressure

$1 \text{ bar} = 10^{+5}\ N/m^2$

$1\,Pa = 1\ N/m^2$

Vertical Component of the surface tensile force

$= \sigma \times \pi d \times \cos\theta$

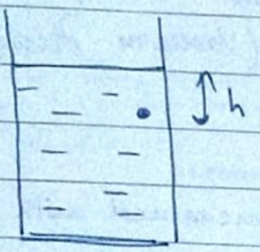
$$\boxed{h = \frac{4\sigma}{\rho g d}}$$

Pascal's law

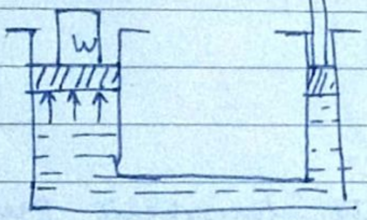
Hydrostatic Law

Capillary Fall = $\left.\dfrac{4\sigma\cos\theta}{\rho g d}\right\}$

⇒ Hydrostatic Law $\rightarrow \rho g h$



Q: A hydrostatic press has a ram of 30cm diameter & a plunger of 4.5cm diameter. Find the weight lifted by the hydraulic press when the force applied on the plunger is 500 N.



$\Rightarrow \left(\dfrac{F}{A}\right)_1 = \left(\dfrac{F}{A}\right)_2 \Rightarrow \dfrac{500}{\pi\cdot\frac{(4.5)^2}{4}} = \dfrac{W}{\pi\cdot\frac{(30)^2}{4}}$

$\Rightarrow \dfrac{500}{(9/2)^2} = \dfrac{W}{900}$

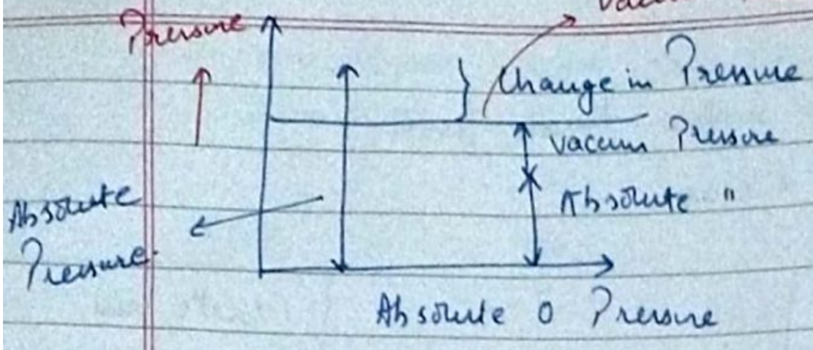
$\Rightarrow \dfrac{500\times 4}{9\times 9} = \dfrac{W}{\cancel{900}\,100} \Rightarrow W = \dfrac{20\times 10^4}{9}$

$= 2.22\times 10^4$ N.

Q: Calculate the pressure due to a column of 0.3m of oil of Specific gravity 0.8.

$P = \rho g h$, $0.8 = \rho/\rho_{water}$, $\rho = 800\ kg/m^3$

Vaccum Pressure (Anything below atmospheric pressure)



⇒ Manometers → Differential, Manual, U-Tube

Mech. Gauges → In Pressure measuring, datum is atmospheric pressure

1.01325 Pa

1.01 MPa $\simeq 1$ MPa

If Pressure $= 0.925$ MPa (Absolute) } below Atmospheric Pressure (Vaccum Pressure)

or -0.125 MPa

Absolute Pressure

It is defined as the Pressure which is measured with ref. to absolute 0 or complete vacuum

Gauge "

__________ with the help of pressure-measuring instrument in which atmospheric pressure is taken as the datum.

Vacuum "

__________ below the atmospheric pressure

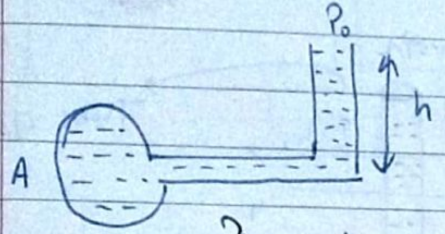
Absolute Pressure = Atmospheric Pressure + Gauge Pressure

Vacuum " = " − Absolute Pressure

Atmospheric pressure at sea level at $15°C$ is 101.3 kPa or 760 mm of Hg or 10.33 m of water

Measurement of Pressure

- Manometer → Simple Manometer, Differential "
  - Simple Manometer → Piezometer, Single Column Manometer, U-Tube Manometer
    - Single Column Manometer → Vertical Single Column, Inclined Single Column
- Mechanical Gauges
  - Diaphragm Pressure Gauge
  - Bourdon Tube " "
  - Dead Wt. " "
  - Ballows " "

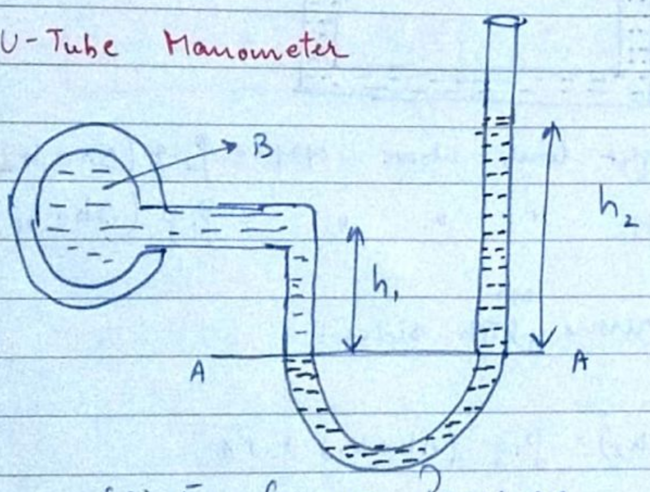
⇒ 

Piezometer

Pressure at A

$$P_A = P_0 + \rho g h$$

⇒ U-Tube Manometer



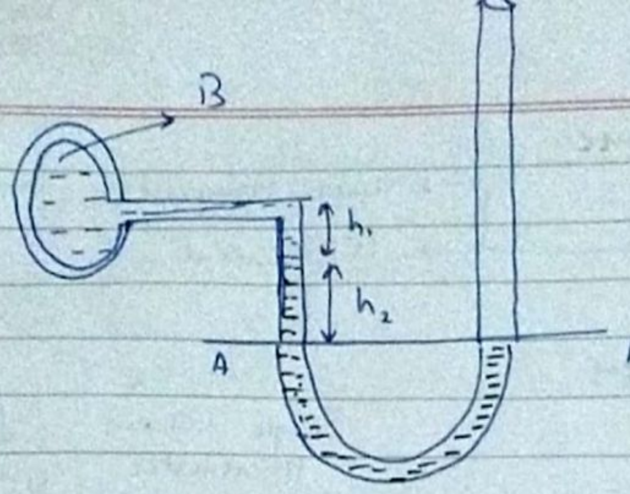
(a) For Gauge Pressure

$$P = \rho g h_1 - \rho g h_2$$

Pressure above AA in the left column $= P + \rho_1 g h_1$ → ①

" " " in the right " $= \rho_2 g h_2$ → ②

Equating ① & ②

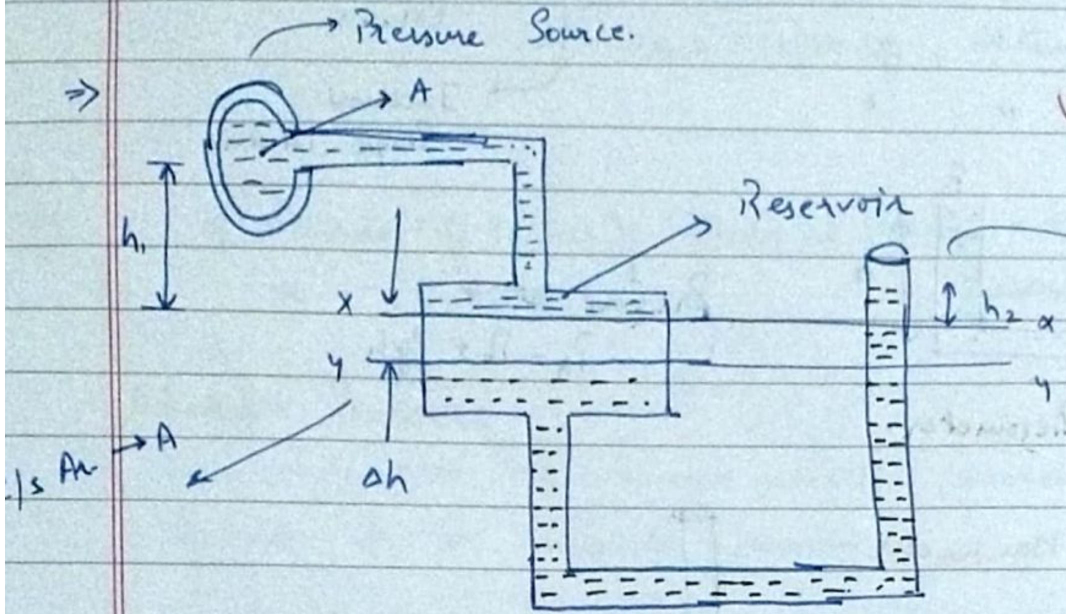


Left Column $= P + \rho_1 g h_1 + \rho_2 g h_2$

Right Column $= 0$

$$\boxed{P = -(\rho_1 g h_1 + \rho_2 g h_2)}$$

(b) For Vacuum Pressure



Vertical Single ~~Vertic~~ Column

Pressure in the right limb above YY $= \rho_2 g(\Delta h + h_2)$

" " " left " " " $= \rho_1 g(\Delta h + h_1) + P_A$

Equating the pressure on both sides:

$$\Rightarrow \rho_2 g(\Delta h + h_2) = \rho_1 g(\Delta h + h_1) + P_A$$

$$P_A = \rho_2 g(\Delta h + h_2) - \rho_1 g(\Delta h + h_1)$$

$$\Rightarrow P_A = \Delta h(\rho_2 g - \rho_1 g) + h_2 \rho_2 g - h_1 \rho_1 g \quad \rightarrow (1)$$

We also have:

Fall ~~ino~~ of the liquid in reservoir ⇒

$$\Delta h \times A = a \cdot h_2 \Rightarrow \boxed{\Delta h = \frac{a}{A} \cdot h_2} \longrightarrow (2)$$

Derⁿ of Bernoulli's Eqⁿ using Euler's Eqⁿ

h_2

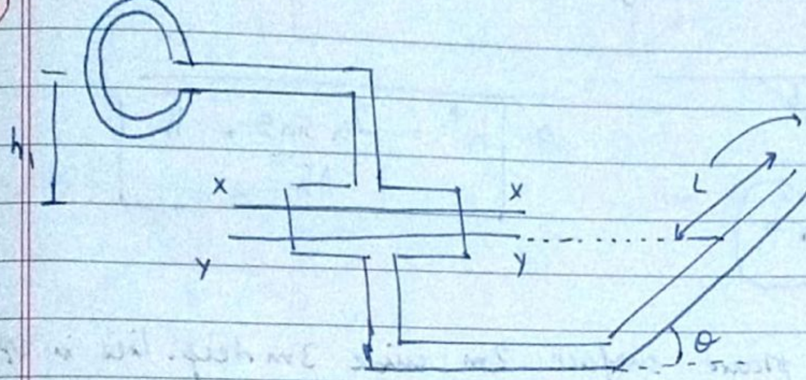
Via ② in ① we get:

$$P_A = \frac{a}{A} \cdot h_2(\rho_2 g - \rho_1 g) + h_2 \rho_2 g - h_1 \rho_1 g$$

As: $A >>> a \Rightarrow \frac{a}{A} \simeq 0$

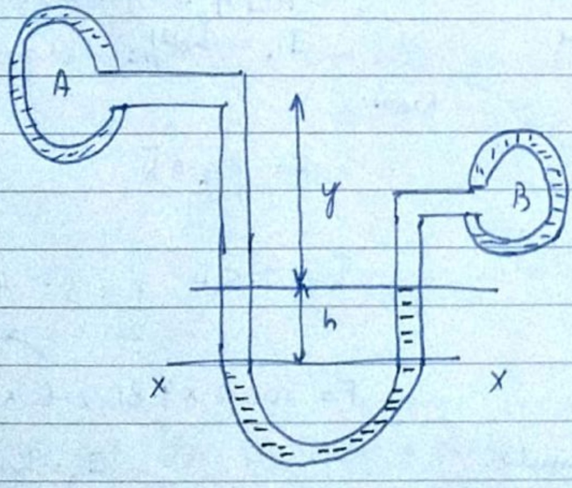
$$\Rightarrow \boxed{P_A = h_2 \rho_2 g - h_1 \rho_1 g} = \text{Final Result}$$

Ⓑ



$$\boxed{P_A = L \sin\theta \, \rho_2 g - h_1 \rho_1 g}$$

Ⓑ Differential Manometer



Left limb $= \rho_1 g (h + x) + P_A$

Right limb $= \rho_g g y + \rho_2 g h_2 + P_B$

$$\boxed{P_A - P_B = (\rho_g - \rho_1) g h + \rho_2 g y - \rho_1 g x}$$

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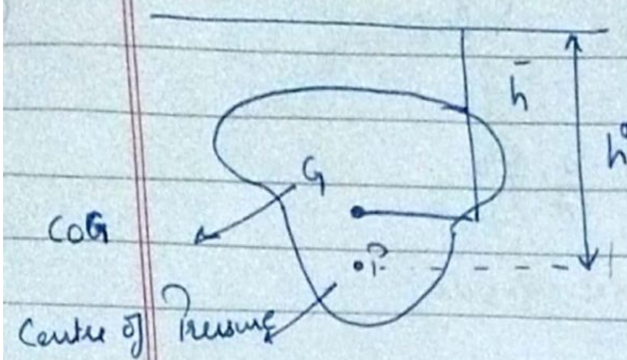
Hydrostatic Forces on Surfaces.



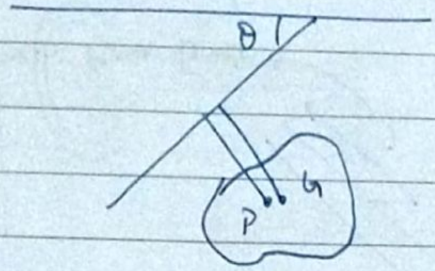
Moment of Inertia. Area

$$\boxed{h^* = \frac{I_{GT} A\bar{h}}{A\bar{h}}}$$

$h^* \to$ h Star

Buoyant force acts on COP

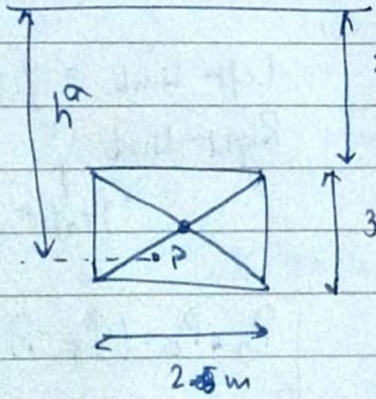
Inclined Surface Submerged in Liquid



$$\Rightarrow \boxed{h^* = \frac{I_G \sin\theta}{A\bar{h}} + \bar{h}}$$

Q: A Rectangular plane surface 2m wide 3m deep. Lies in VP in of water. Determine the total pressure & position of COP on the plane surface when its upper edge is horizontal & 2.5m below water surface.

Ans.



POI of diagonals is the COG

$I_G = bd^3/12$

Now:

$F = \rho g A\bar{h}$

$\bar{h} = 2.5 + \frac{1}{2} \times 3 = 4m$

$F = 1000 \times 9.81 \times \underbrace{6}_{A} \times \underbrace{4}_{\bar{h}}$

Now use h^* formula.

$I_G = \frac{bd^3}{12}$

h^* is 4.9 something.

Q: The fig. shows a tank full of water. Find :→

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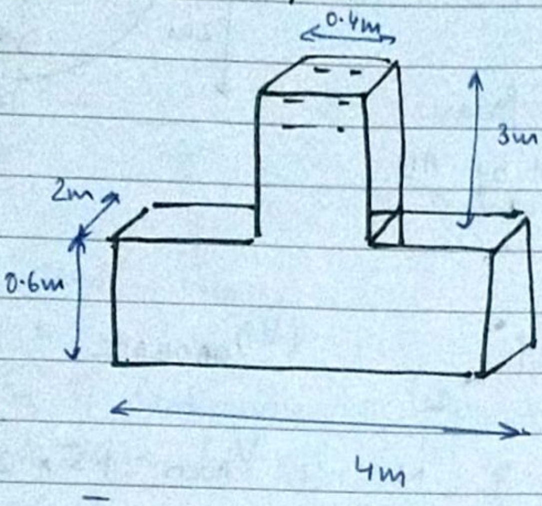
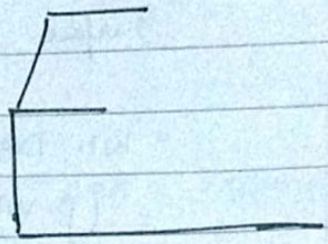
a) T.P. on the bottom surface
b) Wt. of water in the tank
c) Hydrostatic ~~process~~ paradox in result ① & ②.





Ans. a) $F = \rho g A \bar{h}$ → Used to find total pressure at the bottom

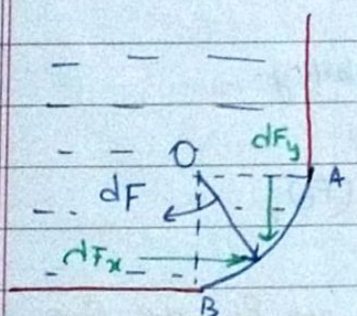
ρ = 1000, g = 9.81, A = 4×2, $\bar{h}$ = 3 + 0.6

b) Wt. of water = $\rho . g .$ (volume of Tank)

$= \rho g \, (l_1 b_1 h_1 + l_2 h_2 b_2)$

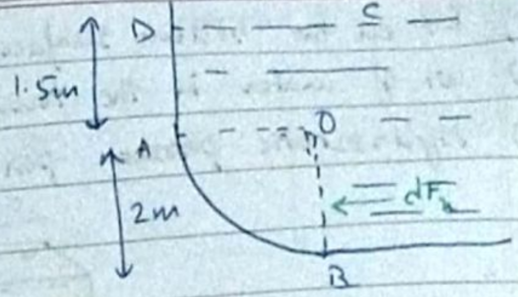
c) Diff. in values of a) & b) is the hydrostatic paradox.

(A) Curved Surfaces submerged in Liquid.



Q: Calculate the horizontal & vertical components of the total force acting on curved surface AB which is of the form of a quarter of circle of radius 2m. Take the width of the surface as unity.

$$\Rightarrow \quad F_x = \rho g A \bar{h}$$



$$= 1000 \times 9.81 \times (OB \times 1) \times 2.5$$

F_y = Wt. of water upto free surface supported by AB

$$= \text{Wt. DABOC}$$

$$= (\rho \cdot \text{volume}) \cdot g$$

$$= 1000 \times (\pi + 3) \cdot g \quad N$$

$$= (1000 \times 9.81) \cdot (6.14)$$

$$(V)_{OBADC} = V_{AOCD} + V_{AOB}$$

$$V_{AOCD} = 1.5 \times 2 \times 1$$

$$V_{AOB} = \left[\frac{\pi \cdot (OB)^2}{4} \middle| \frac{\pi \cdot 2^2}{4} \right] \times 1$$

$$= \pi \ m^3$$

$$\Rightarrow (V)_{DABOC} = (3 + \pi) \, m^3$$

Dynamics of Fluid Flow

$F_g \rightarrow$ Gravity Force

$F_p \rightarrow$ Pressure Force

$F_v \rightarrow$ Force due to viscocity

$F_t \rightarrow$ " " " turbulence

$F_c \rightarrow$ " " " compressibility.

$$F_x = (F_g)_x + (F_p)_x + (F_v)_x + (F_t)_x + (F_c)_x$$

$\Rightarrow$ Ignoring $(F_c)_x$ & keeping the rest : we get Reynold's Eq^n

" $(F_c)_x$ & ~~Fb~~ $(F_t)_x$ " " " : we get Navier Stoke's Eq^n

" " & $(F_t)_x$, F_v " " Bernoulli's Eq^n

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Q: A pipe through which water is flowing is having diameter 20cm & 10cm at the c/s ① & ②. The velocity of water at ① is given as 4 m/s. Find the velocity head at section ① & ②, and also rate of discharge.

Ans
$$\frac{p}{\rho g} + \frac{V^2}{2g} + z = \text{Constant} \quad (\text{Bernoulli's Theorem})$$

$\frac{p}{\rho g}$ → Pressure Head.

$\frac{V^2}{2g}$ → Velocity Head

z → Datum Head.

⇒ & Eqⁿ of Continuity :→ $A_1V_1 = A_2V_2$ → Rate of Discharge

$A_{①} V_{①} = A_{②} V_{②}$

$$\Rightarrow \frac{\pi}{4}\cdot(20cm)^2\cdot 4 = \frac{\pi}{4}\cdot(10)^2\cdot V_{②}$$

$$\Rightarrow 4\cdot 4 = V_2 \Rightarrow V_2 = 16 \text{ m/s} \longrightarrow ①$$

Velocity Head at ① ⇒ $\frac{V_{①}^2}{2g} = \frac{4^2}{2(9.81)}$ & lly at ②

Bernoulli's Eqⁿ for Real Fluids

$$\boxed{\frac{p_1}{\rho g} + \frac{v_1^2}{2g} + z_1 = \frac{p_2}{\rho g} + \frac{V_2^2}{2g} + z_2 - h_L}$$

h_L → Loss of Head.

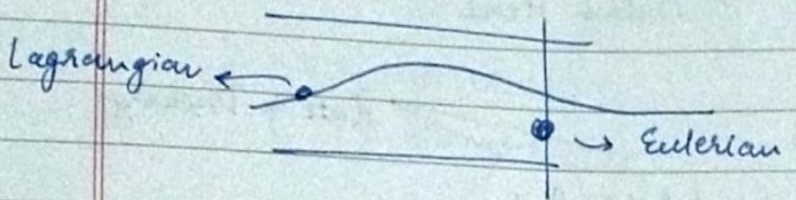
(A) ⇒ Fluid Motion
- → Lagrangian Method
- → ~~Eaclsdan~~ Eulerian Method.

Types of Fluid Flow
- ↳ Steady & unsteady flow
- ↳ Uniform & Non Uniform "
- ↳ Laminar & Turbulent
- ↳ Compressible & incompressible
- ↳ Rotational & Irrational Value
- ↳ One, Two & 3-Dimensional Flow

Definition.

a) Langragian Method : A single fluid particle is followed during its motion & its velocity, accⁿ & density are computed.

b) Eulerian Method : Velocity, Pressure & density are defined at a pt. in a fluid flow.



→ Steady Flow is that type of flow in which the fluid characteristics like velocity, pressure & density at a pt. do not change with time.

$$\left(\frac{\partial v}{\partial t}\right)_{x_0,y_0,z_0} = 0 \quad , \quad \left(\frac{\partial p}{\partial t}\right)_{x_0,y_0,z_0} = 0 \quad \& \quad \left(\frac{\partial \rho}{\partial t}\right)_{x_0,y_0,z_0} = 0$$

For unsteady flow, equality doesn't hold. $\neq 0$

→ Uniform Flow : is that type of flow in which velocity at any given time does not change w.r.t space

$$\Rightarrow \left(\frac{\partial v}{\partial s}\right)_{t=\text{constant}} = 0$$

For Non-uniform, flow equality doesn't hold.

→ Laminar : ______ in which the fluid particles move along well defined path or ~~straight lines~~ streamlines. Thus particles move in laminars or layer gliding smoothly over the adjacent layer.

* Darcy, Chezy's Formula

Assignment: → Flow through pipes
2 Types of losses → Major, → Minor (5) } Causes & Formulae. → ke 2 formulae *

⇒ Turbulent : ______ in which the fluid particles move in a zig zag way. Due to this zig-zag motion, the eddie's formations takes place which are responsible for high energy loss.

For a pipe flow, ~~each~~ the type of flow is determined by a non-dimensional no. expressed as:

Re/N = Reynold's No. $= \frac{VD}{\nu \text{ (viscosity)}}$ → D: Diameter of Pipe; viscosity → Kinematic

$Re < 2000$ – Laminar
> 4000 → turbulent
$2000 < Re < 4000$
Laminar or Turbulent

α —— α_c —— α

22/8/19.

Energy Losses

(due to friction)

Major Energy Losses
1. Darcy-Weibash Theorem
2. Chezy's Formula

Minor Energy Losses
1) Sudden Expansion in pipe
2) " Contraction " "
3) Bend in pipe
4) pipe fitting
5) Obstruction in pipe.

Darcy's-Weishbach

$$h_f = \frac{4fLV^2}{d \cdot 2g} \quad ; \quad f = \frac{16}{R_e} \text{ for } Re < 2000$$

$$= \frac{0.079}{R^{1/4}} \quad , \quad Re \in [4000, 10^6]$$

Chezy's Formula.

$$h_f = \frac{f'}{\rho g} \times L V^2 \times \frac{P}{A}$$

$\Rightarrow$ $\frac{A}{P} = \frac{\text{Area of Flow}}{\text{perimeter}}$ = Hydraulic Mean Depth

$$m = \frac{A}{P} = \frac{\pi d^2/4}{\pi d} = \frac{d}{4}$$

Thus: $\frac{P}{A} = \frac{1}{m}$

Substituting in above eq^n we have:

$$\Rightarrow h_f = \frac{f'}{\rho g} \times L \cdot V^2 \times \frac{1}{m}$$

or: $V^2 = \frac{h_f \times \rho g \times m}{f' L}$ $\Rightarrow V = \sqrt{\frac{h_f \times \rho \cdot g \cdot m}{f' L}}$

$$\Rightarrow V = \underbrace{\sqrt{\frac{\rho g}{f'}}}_{C} \times \sqrt{\frac{h_f \cdot m}{L}}$$

& $\frac{h_f}{L} = i$ Loss of Head per unit length of pipe

$$\Rightarrow \boxed{V = C\sqrt{i \cdot m}}$$

Q: Find the head lost due to friction in ~~two~~ pipe of diameter = 300 mm & length 50 m through which water is flowing at a velocity of 3 m/s, using
(i) Darcy (ii) Chezy's Formula for $C = 60$
Take $\mu_K = 0.01$ Stoke.

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$\rightarrow ①$

Ans. $h_f = \frac{4fLv^2}{d \cdot 2g}$, $d = 300mm$
$= 300 \times 10^{-3}$ m
$L = 50m$
$V = 3 m/s$

$Re = \frac{VD}{\nu}$

$= \frac{\cancel{3e} \; 3 \times 300 \times 10^{-3}}{0.01 \text{ Stoke}}$

$0.01 \text{ stoke} = 0.01 \frac{cm^2}{s}$

$= \frac{0.01 \times (10^{-2})^2}{1 s}$

$= 0.01 \times 10^{-4}$

$\Rightarrow Re = \frac{3 \times 300 \times \cancel{10^{-3}}}{1 \times \cancel{10^{-6}}} 10^3$

$= 9 \times 10^5$

Thus: formula for: $f = \frac{0.079}{(Re)^{1/4}}$

$\Rightarrow f = \frac{0.079}{(9 \times 10^5)^{1/4}}$

$\Rightarrow f = \frac{0.079}{30.80} = 0.0025 \longrightarrow Ⓐ$

Via Ⓐ in ① we get:

$\Rightarrow h_f = \frac{4 \times 0.0025 \times 50 \times 3^2}{300 \times 10^{-3} \times 2 \times 9.81}$

$= 0.076$ = Ans

(ii) $h_f =$ $v^2 = c^2 im$, $m = d/4$, $i = \frac{h_f}{L}$

$\Rightarrow v^2 = (60)^2 \times \frac{h_f}{L} \times \frac{d}{4} \Rightarrow 9 = 36 \times 10^2 \times \frac{h_f}{50} \times \frac{0.3}{4}$

$\Rightarrow h_f = \frac{5\cancel{0} \times 1\cancel{0}}{1\cancel{0}^2 \times 3} = 1.66$ = Ans.

$\rightarrow$ Chezy's Formula

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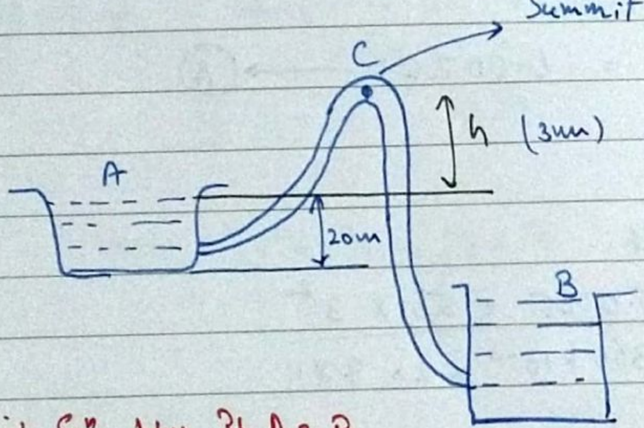
Flow Through Syphon.

Syphon : is a long bent pipe which is used to transfer liquid from a reservoir at higher location to a reservoir at lower position, when the 2 " s are seperated by a hill or high level ground.

$P_c < P_{atm.}$

Q: A syphon of diameter 200mm ~~collects~~ connects 2 reservoirs having a cliff in elevation of 20m, length of syphon is 500m & the summit is 3m above the water level in upper reservoir.
The length of pipe from above reservoir to summit is 100m.
Determine discharge ~~at~~ through the syphone & pressure . Neglect minor energy losses.

Ans.



Applying Bernoulli's Eqn b/w Pt. A & B

$$\frac{P_A}{\rho g} + \frac{V_A^2}{2g} + Z_A = \frac{P_B}{\rho g} + \frac{V_B^2}{2g} + Z_B + \text{Loss of Head due to friction}$$

$$0 + 0 + Z_A = 0 + 0 + Z_B + h_f$$ (Minor losses have been ignored)

$\therefore (Z_A - Z_B = 20m)$

$$Z_A - Z_B = h_f = \frac{4fLV^2}{d \times 2g} \Rightarrow V = 2.80 m/s \longrightarrow (1)$$

Entrance + Darcy + loss due to entrance + Darcy

Sudden Expansion of pipe + Darcy + loss due to exit
contraction

Applying Bernoulli's Eqⁿ at A & C,

$$\Rightarrow \frac{P_A}{\rho g} + \frac{V_A^2}{2g} + Z_A = \frac{P_c}{\rho g} + \frac{V_c^2}{2g} + Z_c + h_f$$

$$0 \quad 0 \quad 0 = \left(\frac{P_c}{\rho g}\right) + \frac{V^2}{2g} + h_f \rightarrow (2) \text{ Use (1) here}$$

Ans

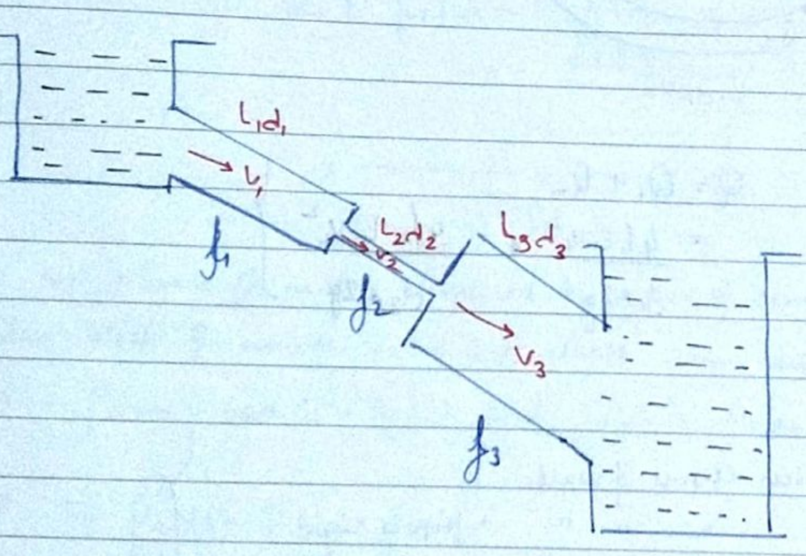
$$\Rightarrow \frac{P_c}{\rho g} = -7.399 \text{ m of Hg.}$$

Ans → (ii) Q = ~~Velocity~~ Discharge Through syphon

$= (2.80)\ \frac{\pi}{4} \times$ ___

(A)

Flow Through Pipes in Series.



$$H = \frac{0.5\,V_1^2}{2g} + \frac{4f_1 L_1 V_1^2}{d_1 \times 2g} + \frac{0.5\,V_2^2}{2g} + \frac{4f_2 L_2 V_2^2}{d_2 \times 2g} + \frac{(V_2 - V_3)^2}{2g}$$

$$+ \frac{4 f\, l_3 V_3^2}{d_3 \times 2g} + \frac{V_3\, g}{2g}.$$

If minor losses are neglected

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(#) Derivation of Dupit's Eqn

(#) Flow Through ~~Bdo~~ Branched pipe (Numerical)

(#) Pour Transmission Pipes derivation

•) Flow Through Nozzle

•) Water-hammering Pipes

$$\boxed{h = \frac{4f_1 L_1 V^2}{d_1 \times 2g} + \frac{4f_2 L_2 V_2^2}{d_2 \times 2g} + \frac{4f_3 L_3 V_3^2}{d_3 \times 2g}}$$

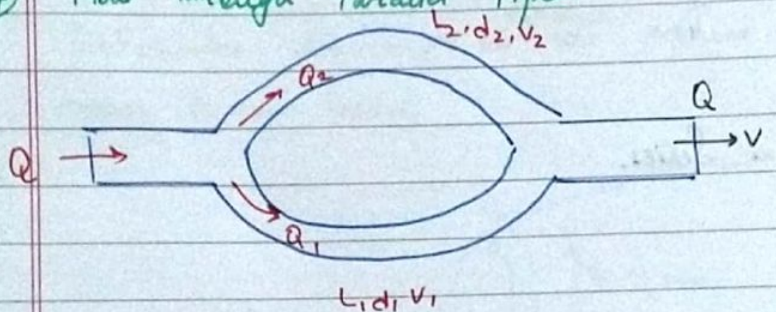
(#) Equivilent Pipe :→ This is the defined as the pipe of uniform diameter having loss of head & discharge equal to — of a compound pipe consisting of several pipes of different length & diameter.

$$\boxed{\frac{L_1}{d_1^5} + \frac{L_2}{d_2^5} + \frac{L_3}{d_3^5} = \frac{L}{d^5}}$$

↓

Dupuit's Equation.

(#) Flow through Parallel Pipes



$$Q = Q_1 + Q_2$$

$$= \frac{4f_1 L_1 V_1^2}{d_1 \times 2g} = \frac{4f_2 L_2 V_2^2}{d_2 \times 2g}$$

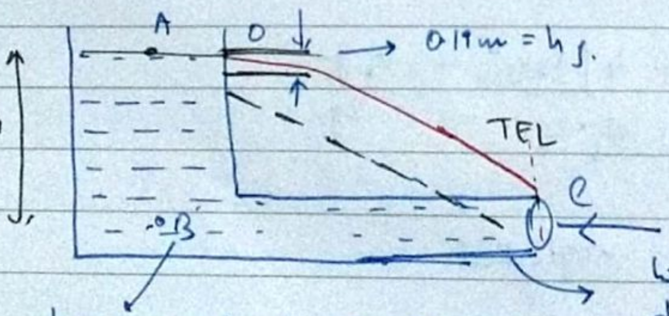
Water Hammer

↳ Sudden Closure of wall

↳ " " " " → pipe is rigid

↳ ———— → " " elastic.

⇒



& Just have to plot lines.

o) 1 Solved numerical on α concept | HGL | TEL

o) Derv^n $\rightarrow$ $T = 2\pi\sqrt{\frac{k^2}{GM \cdot g}}$

Meta-Centric Distance

\# Hydraulic Gradient Line

It is defined as the line which gives the sum of pressure head & datum head of a flowing fluid in a pipe w.r.t. some ref. line

\# Total Energy line

——— sum of pressure head, datum head & kinetic head of a flowing fluid in a pipe w.r.t. some reference line

α $\rightarrow$ Soln to this.

Pipe attached to reservoir's centre line is the ref line.

h_f = Head lost during the entrance of pipe

$$= \frac{0.5 \cdot v^2}{2g} = \frac{0.5 \times 2.734}{2 \times 9.81} = 0.19\,m.$$

Head lost due to friction $= \frac{4fLV^2}{d \times 2g} = 3.423\,m.$

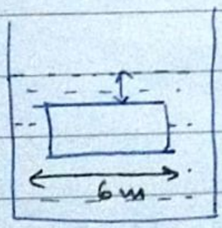
α ——— $\ast$ ——— α

29/8/19

Q: Find the volume of water displaced & position of centre of buoyancy for a wooden block of width 2.5m & of depth 1.5m, when it floats horizontally in water. ρ_{wood} = ~~600~~ 650 kg/m^3 & its length is 6m.

Ans:

$F = \rho g h(A) = P.A.$



$(F)_B$ = Wt. of Fluid Displace

$\Rightarrow \rho V_w \cancel{g} = V_{Fluid} \cdot \rho_{Fluid} \times \cancel{g}$

$\Rightarrow 650\,\frac{kg}{m^3} \times 6 \times 1.5 \times 2.5\ m^3 = V_{Fluid} \times 1000\ kg/m^3$

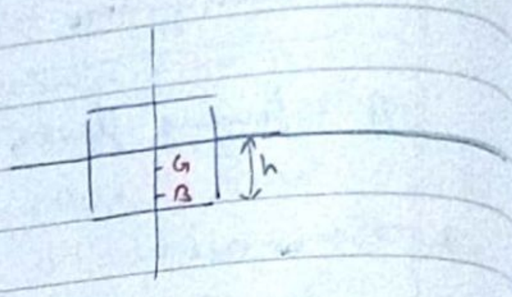
$V_{Fluid} = 14.625\ m^3 = 14625$ litres

$F_{Buoyancy}$ = wt. of Fluid Displaced

P.A

$\Rightarrow$ $\rho_F \cdot g \cdot h \cdot A = g \times \rho_F$

Sir's Answer



(#) Buoyancy & Flotation

Volume of block = $l \times b \times h$

$= 22.50\ m^3$

Weight = $\rho g V = 143471$ N.

For Equilibrium

Weight of water displaced = wt. of block

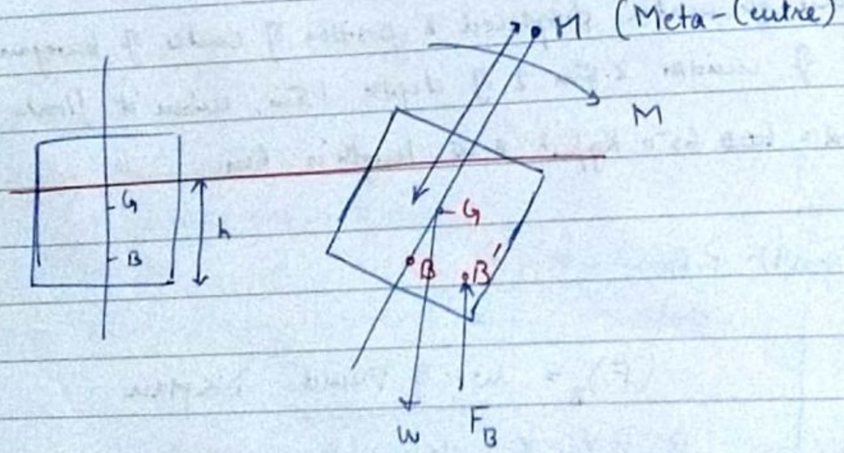
volume of water displaced $= \dfrac{14371}{9.8 \times 1000} = 14.625\ m^3$.

$\Rightarrow$ Volume of wooden block in water = Volume of water displaced

$\Rightarrow$ $2.5 \times h \times 6 = 14.625$

$h = 0.975m$

Centre of buoyancy $= \dfrac{0.975}{2} = 0.4875\ m$



Meta-Centre

It is defined as a pt. about which body starts oscillating when tilted by small angle.

It is also defined as a pt. at which line of action of force of buoyancy will meet the normal axis of the body.

Formula:

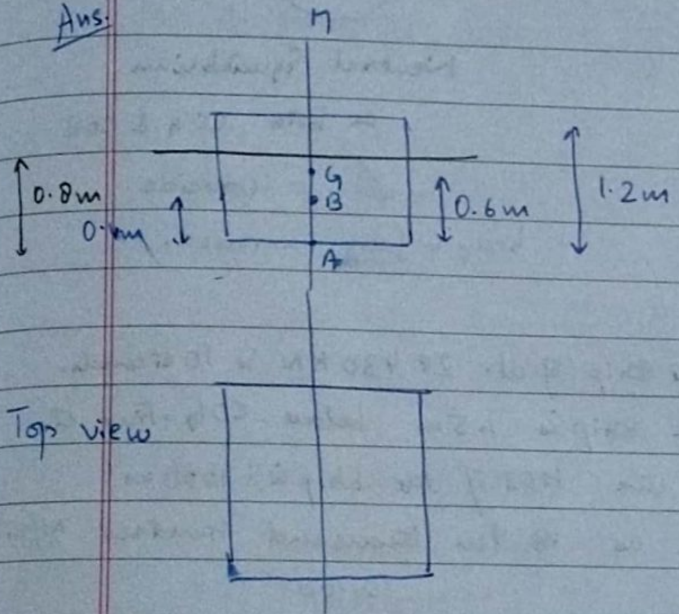
$$\boxed{MG = \text{Meta-centric distance} = BM - BG = \frac{I}{\forall \text{ (Vol. of body submerged in liq.)}} - BG}$$

I for box: $\frac{bd^3}{12}$

" Cylinder: $\frac{\pi d^4}{64}$.

Q: A rectangular body, $l = 5m$, width = 3m & ht. 1.20m, is immersed in H_2O depth of immersion = 0.8m. If CoG is 0.6m from the bottom of the sea-level. Find meta-centric ht. $\rho_{sea\ water} = 1025\ kg/m^3$

Ans:



d = depth

$$I = \frac{5 \times 3^3}{12} \left(\frac{bd^3}{12}\right) = \frac{45}{4}\ m^4$$

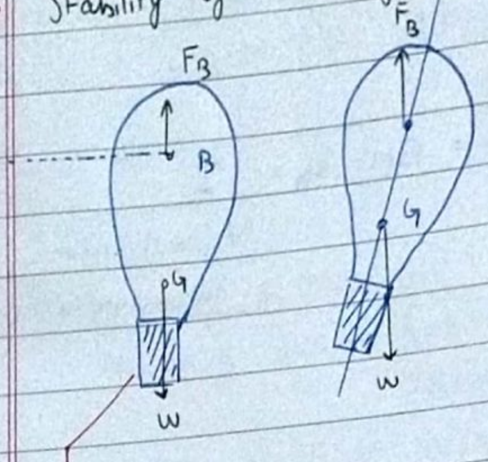
$$V_{Body\ submerged} = 3 \times 0.8 \times 5 = 12 m^3$$

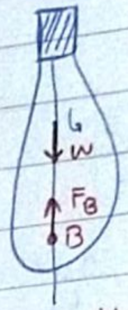
$$\left.\begin{aligned} BG &= AG - AB \\ &= 0.6 - 0.4 \\ &= 0.2\ m \end{aligned}\right\} =$$

$$\Rightarrow \boxed{GM = \frac{45}{4} \times \frac{1}{12} - 0.2 = 0.7357\ m}$$

Stability of Submerged Body :

⇒ Stability of Submerged Bodies



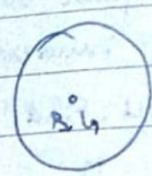


Unstable Equilibrium

Stable Equilibrium

Shaded part is heavy so COG is below towards COB.

Due to couple body comes to its original position.



Neutral Equilibrium as both COG & COB coincide

body is fully immersed.

Q: The time period of rolling of a ship of wt. 29 430 kN is 10 seconds. The centre of buoyancy of the ship is 1.5 m below COG. Find the radius of gyration of ship if the MOI of the ship is 1000 m^4. Take specific wt of sea water as ten thousand hundred N/m³.

10,100

Ans.

$$T = 2\pi\sqrt{\frac{k^2}{GM \times g}}, \quad T^2 = \frac{4\pi^2 \cdot k^2}{GM \times g}$$

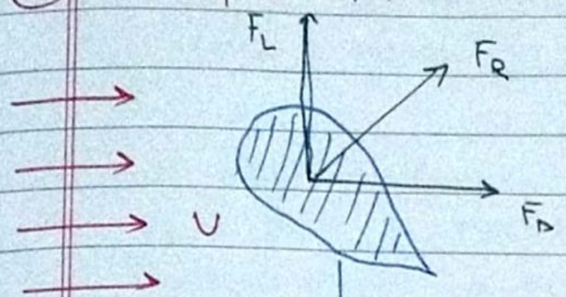
$$\Rightarrow (10)^2 = \frac{4\pi^2 \cdot k^2}{GM \times g}$$

$GM = BM - BG$

$= \frac{I}{\forall} - BG$

$\forall = \frac{\text{Wt. of ship}}{\text{Specific gravity of sea water}}$ → Volume of water displaced.

A1 DRAG & LIFT



Drag force

C_D

$$F_D = C_D\, A\, \frac{\rho V^2}{2}$$

Lift Force,

$$F_L = C_L\, A\, \frac{\rho V^2}{2}$$

$$F_R = \sqrt{F_D^2 + F_L^2}$$

→ Stationary Body

Aerofoil section

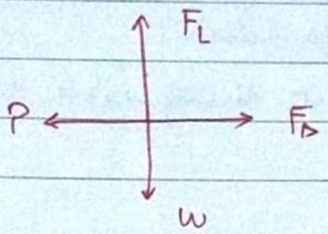
c/s of the wing.

Aerodynamic Shape: Should generate Max. lift force & minimum drag.

A. Ar. projected to moving fluid

C_L = Coeff. of Lift.

FBD of Plane:



Angle of Attack: ∠ made by the wing against the fluid.

α —— α —— α

THERMODYNAMICS

Macroscopic Approach: Classical Thermodynamics

Microscopic " : Statistical Thermodynamics

→ Not studying every individual particle's heat content.

System